

Pension Run

Kai-Jyun Wang

- Why people seldom annuitize the pension?
 - Retire later to accumulate more wealth.
 - Retire earlier to withdraw more pension.
 - Reform Uncertainty.
 - Run on the pension system.
- Related literature:
 - **Pension reform:** Bi and Zubairy (2023); Daminato and Padula (2024); Delalibera, Ferreira, and Machado Parente (2026); French and Jones (2011); Kitao (2018).
 - **Bank run:** Goldstein and Pauzner (2005); He and Xiong (2012).

Background

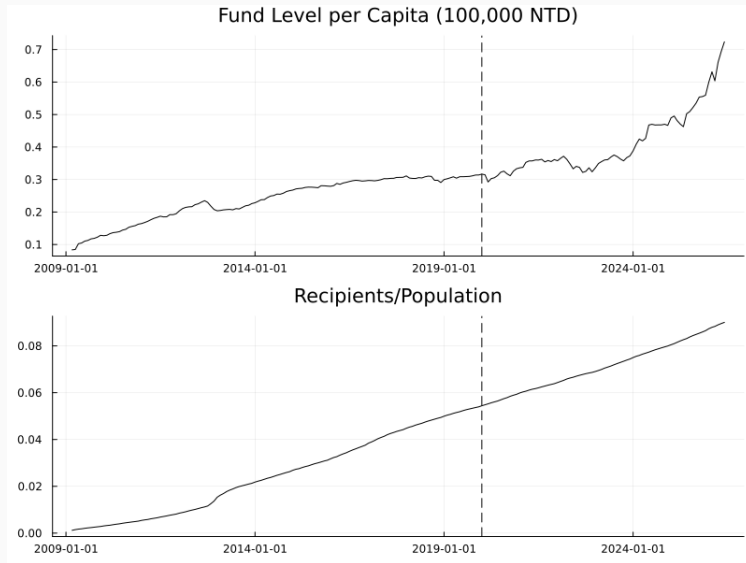
- Since 2009, one with insurance records before 2009 can choose between lumpsum and monthly scheme.
- For a recipient with average monthly insured salary NTD 45,800 with 30 years of working retired at standard retirement age and lives 12 years after retirement.
 - Lumpsum: 2,061,000 NTD.
 - Monthly: 2,724,504 NTD. (2% annual rate)
- Reform information shocks:
 - Reform committee established in 2012.
 - Reform on Civil Servant Retirement Pension in 2017.
 - External finance from the government since 2020.

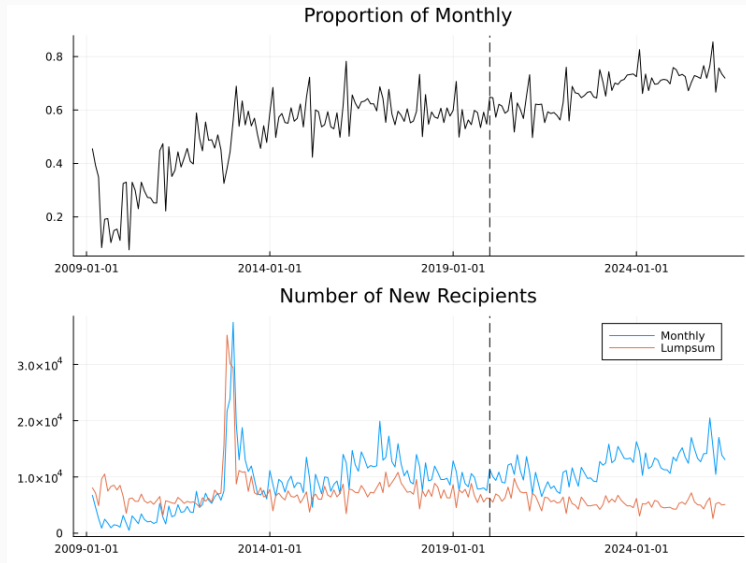
Data

- **Official monthly sources (since 2009):**
 - Bureau of Labor Funds: fund balance and investment return.
 - Bureau of Labor Insurance: stocks of monthly recipients and new lump-sum claims.
 - Ministry of the Interior: population and mortality at ages 60+.
- BLI reports new monthly recipients (first payments) only from May 2022. Before then, reconstruct entries from the stock of monthly recipients:

new monthly recipients = change in the recipient stock + estimated deaths
+ average correction from the overlap period.

Estimated deaths use the monthly mortality rate implied by annual mortality among Taiwan residents aged 60+; negative estimates are set to zero.





Model

Model Overview

- Continuous time and an infinite horizon.
- A continuum of workers chooses once, at retirement age T , between a lump-sum benefit ℓ and a monthly pension p .
- The pension fund earns a stochastic return and receives payroll tax revenue.
- Benefits stop when the fund balance reaches zero.
- Retirees' choices affect fund outflows and, consequently, the value of the monthly pension.

Population

Let $g(t, s)$ denote the population density at age s :

$$\frac{\partial g}{\partial t} = - \underbrace{\frac{\partial g}{\partial s}}_{\text{aging}} - \underbrace{m(s)g}_{\text{mortality}} + \underbrace{\int b(s)g(t, s) ds}_{\text{birth}} \delta_0.$$

Assume $b(s) = b$ and $m(s) = m\mathbf{1}\{s \geq T_m\}$. In a steady-age distribution, N_t grows at rate n and

$$\hat{g}(s) = \begin{cases} be^{-ns}, & s < T_m, \\ be^{-ns}e^{-m(s-T_m)}, & s \geq T_m, \end{cases}$$

where

$$1 = b \frac{1 - e^{-nT_m}}{n} + be^{-nT_m} \frac{1}{n + m}.$$

► View population density

Pension System

Let q_t be the share of new retirees choosing the monthly pension and Q_t the stock of monthly-benefit recipients. For $T \geq T_m$,

$$dQ_t = \left[\underbrace{q_t \hat{g}(T) N_t}_{\text{new retirees}} - \underbrace{m Q_t}_{\text{mortality}} \right] dt.$$

The pension fund balance evolves according to

$$dB_t = \left[\underbrace{r B_t}_{\text{interest}} + \underbrace{\tau y N_t \int_{T_w}^T \hat{g}(s) ds}_{\text{pension tax}} - \underbrace{(1 - q_t) \ell N_t \hat{g}(T)}_{\text{lump sum}} - \underbrace{p Q_t}_{\text{monthly}} \right] dt + \sigma B_t dZ_t.$$

The system is bankrupt when B_t first reaches zero.

Detrended State Dynamics

Define fund balance and recipients per capita by $\hat{B}_t = B_t/N_t$ and $\hat{Q}_t = Q_t/N_t$. Then

$$d\hat{B}_t = \left[\underbrace{(r - n)\hat{B}_t}_{\text{net return}} + \underbrace{\tau y \int_{T_w}^T \hat{g}(s) ds}_{\text{pension tax}} - \underbrace{(1 - q_t)\hat{g}(T)\ell}_{\text{lump sum}} - \underbrace{p\hat{Q}_t}_{\text{monthly}} \right] dt + \sigma \hat{B}_t dZ_t,$$
$$d\hat{Q}_t = \left[\underbrace{q_t \hat{g}(T)}_{\text{new retirees}} - \underbrace{(m + n)\hat{Q}_t}_{\text{mortality}} \right] dt.$$

- More lump-sum claims reduce the fund immediately.
- More monthly claims raise future liabilities through $\hat{Q}_t$.

Individual Pension Choice

A retiring individual chooses $d \in \{\ell, p\}$ with utility

$$u_{it}(d) = \begin{cases} \alpha + \log(\ell) + \epsilon_i(\ell), & d = \ell, \\ \log(V(\hat{B}_t, \hat{Q}_t)) + \epsilon_i(p), & d = p. \end{cases}$$

The value of monthly benefits is

$$V(\hat{B}, \hat{Q}) = \mathbb{E} \left[\int_0^{T_{\hat{B}}} e^{-(\rho+m)s} p \, ds \mid \hat{B}, \hat{Q} \right].$$

With i.i.d. standard Gumbel ϵ_i , the monthly-benefit recipient share is

$$q(\hat{B}, \hat{Q}) = \frac{1}{1 + \exp \left(\alpha + \log(\ell) - \log(V(\hat{B}, \hat{Q})) \right)}.$$

Markov Equilibrium

The equilibrium objects are the monthly-pension choice $q(\hat{B}, \hat{Q})$ and its value $V(\hat{B}, \hat{Q})$ such that

1. the state dynamics induced by q ;
2. the value V implied by the resulting bankruptcy time; and
3. the individual choices induced by V , which aggregate to q .

Value Function and Numerical Solution

The monthly-pension value solves the nonlinear HJB equation

$$(\rho + m)V = p + \mathcal{A}V, \quad V(0, \hat{Q}) = 0,$$

where

$$\mathcal{A}f = \mu_B \frac{\partial f}{\partial \hat{B}} + \mu_Q \frac{\partial f}{\partial \hat{Q}} + \frac{1}{2} \sigma^2 \hat{B}^2 \frac{\partial^2 f}{\partial \hat{B}^2}.$$

- Nonlinearity arises because μ_B and μ_Q depend on q , which depends on V .
- At a sufficiently large upper boundary, impose $\partial V / \partial \hat{B} \approx 0$.
- Solve by finite differences and successive iteration, equivalent to semi-implicit scheme with $dt = \infty$.

► [View algorithm](#)

Results

Calibration

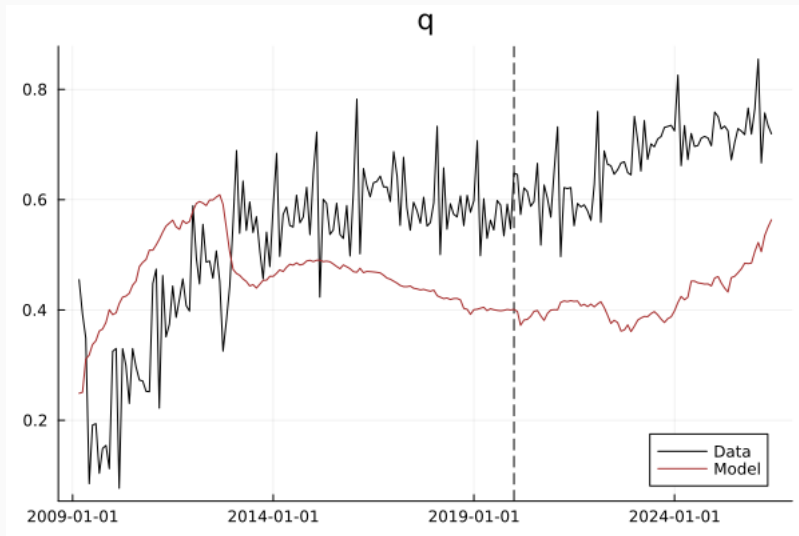
Parameter	Description	Value	Target
<i>Panel A: External Calibration</i>			
b	Birth rate	0.00462	Total fertility rate
m	Mortality rate	0.06	
T_w	Working age	25	
T_r	Retirement age	61	Average retirement age
T_m	Age of mortality exposure	60	
r	Pension fund return rate	0.057	Mean annualized pension fund return
τ	Pension tax	0.075	Pension tax
y	Income	5.268	Highest monthly insured salary
ℓ	Lump-sum pension	19.75	Transfer based on 30 years of working
p	Monthly pension	2.449	Pension based on 30 years of working

Note: The monetary unit in the model is 100,000 NTD.

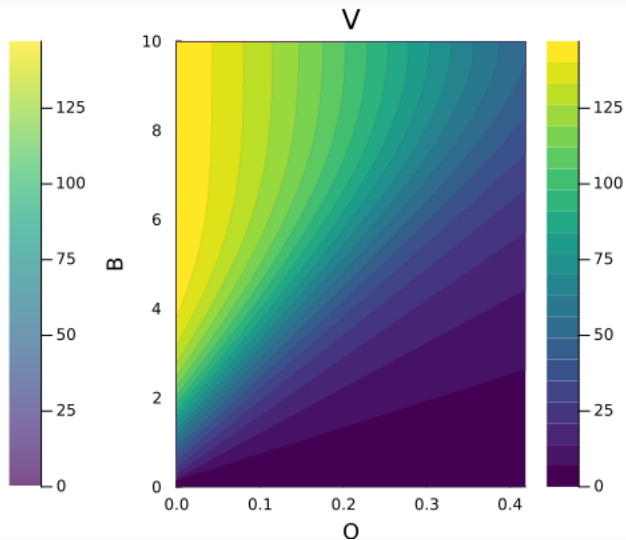
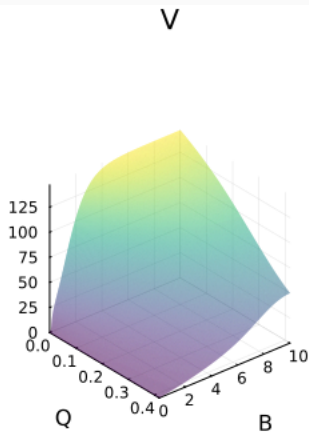
Parameter	Description	Value	Target
<i>Panel B: Internal Calibration</i>			
σ	Pension fund volatility	0.1002	$SD\left(\frac{\hat{B}_{t+\Delta} - \hat{B}_t - \mu_B(\hat{B}_t, \hat{Q}_t, \theta)\Delta}{\hat{B}_t\sqrt{\Delta}}\right) = \sigma$
α	Preference for lump-sum scheme	-1.205	$E\left[(q - q(\hat{B}_t, \hat{Q}_t, \theta))^2\right]$
ρ	Subjective discount rate	-0.0599	$E\left[(q - q(\hat{B}_t, \hat{Q}_t, \theta))^2\right]$

Note: The monetary unit in the model is 100,000 NTD.

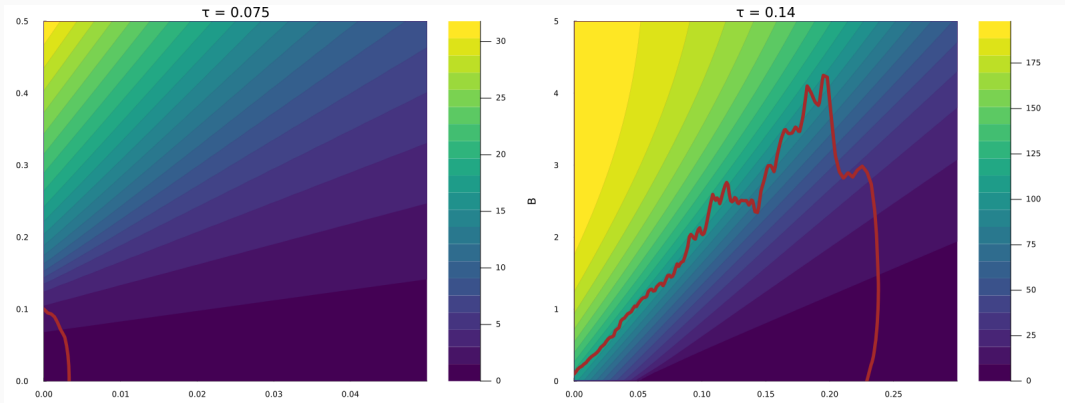
Validation



Estimated V



Externalities



- **Liquidity Channel:**

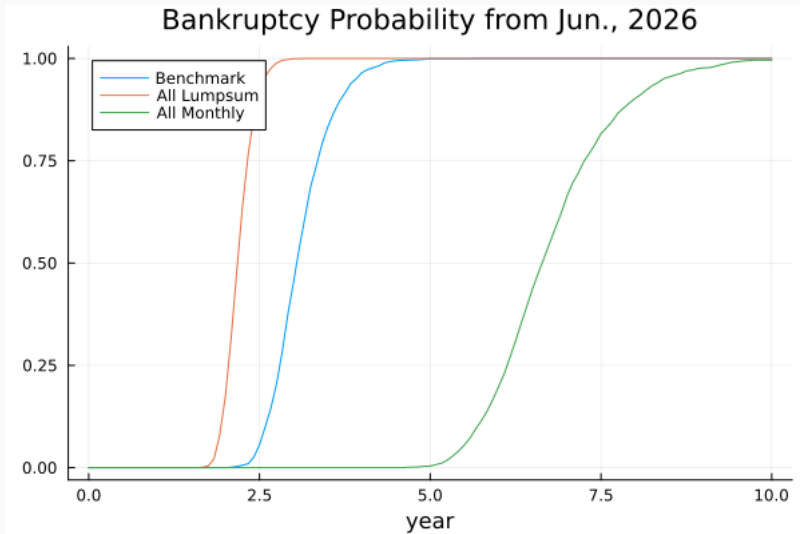
- Weak fundamental → near bankruptcy → lumpsum dominates → accelerate bankruptcy

- **Solvency Channel:**

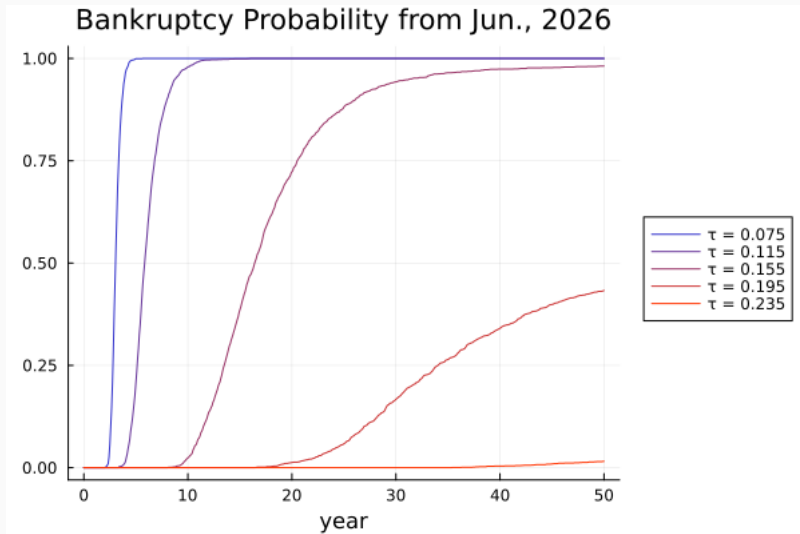
- Stronger but still insolvent fundamental → later bankruptcy → initially monthly dominates → add stress on fund balance

- Experiments
 - Simulate from 2020.
 - No monthly option or lumpsum option ($\alpha \rightarrow \pm\infty$).
 - Raise pension tax rate.
 - Cut benefits.

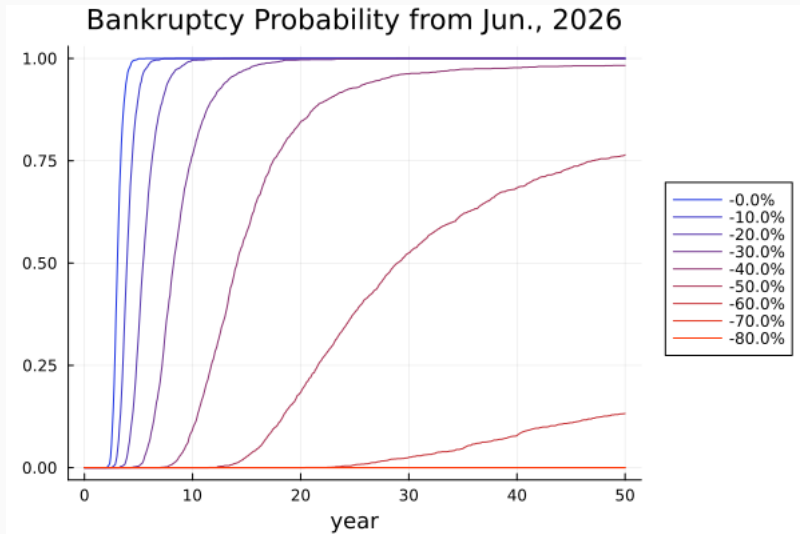
Bankruptcy Probabilities



Bankruptcy Probabilities for Different τ



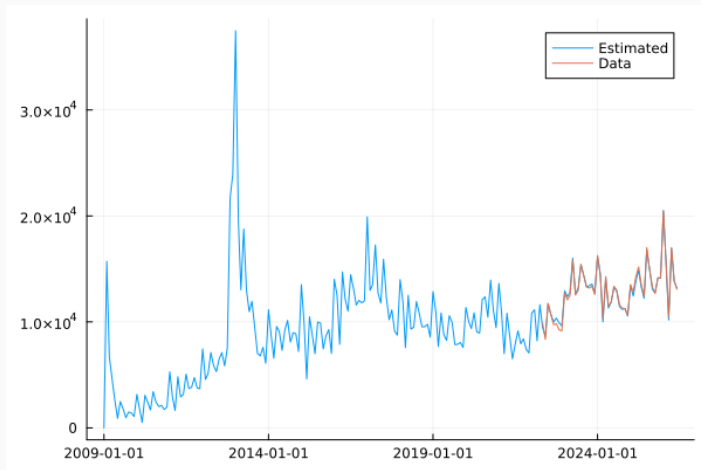
Bankruptcy Probabilities for Cutting Benefits



- Take away:
 - The more generous monthly scheme creates the solvency issue.
 - Raising tax rate to 23.5% or cutting benefits by 70% reduces the bankruptcy probabilities within 50 years near 0.
- To do:
 - Relax the mortality assumption.
 - Belief process.
 - Endogenous retirement.
 - Boom period in return.

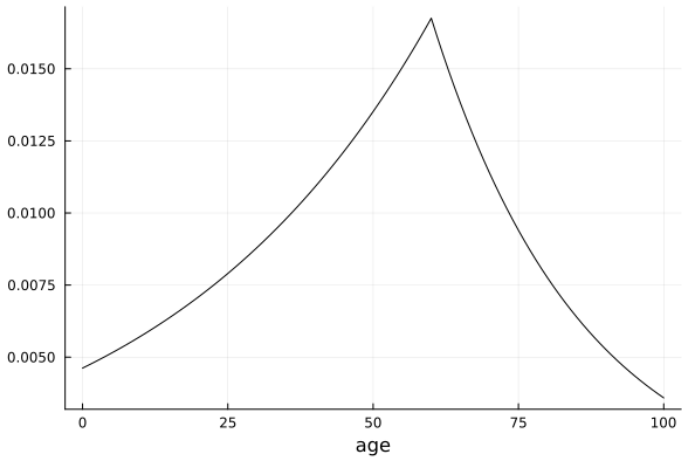
Appendix

Reconstruction Validation



[◀ Back to data reconstruction](#)

Population Density



◀ Back to population

Successive Iteration Algorithm

Require: Initial guess $V^{(0)}$, tolerance $\epsilon > 0$, maximum iterations K

1: Set $k \leftarrow 0$

2: **repeat**

3: Given $V^{(k)}$, compute

$$\mu_B^{(k)} = \mu_B(B, Q, V^{(k)}(B, Q)), \quad \mu_Q^{(k)} = \mu_Q(B, Q, V^{(k)}(B, Q))$$

and the infinitesimal generator

$$\mathcal{A}^{(k)} = \mu_B^{(k)} \frac{\partial}{\partial B} + \mu_Q^{(k)} \frac{\partial}{\partial Q} + \frac{1}{2} \sigma^2 B^2 \frac{\partial^2}{\partial B^2}.$$

with adjustments to the boundary nodes.

4: Solve the linear system

$$\left((\rho + m)I - \mathcal{A}^{(k)} \right) V^{(k+1)} = \mathbf{b}, \quad \mathbf{b} = \rho \operatorname{vec} \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 1 \\ \vdots & \ddots & \ddots & \vdots \\ 1 & \cdots & 1 & 1 \end{pmatrix}.$$

5: Compute the error

$$e^{(k+1)} \leftarrow \left\| V^{(k+1)} - V^{(k)} \right\|_{\infty}.$$

6: $k \leftarrow k + 1$

7: **until** $e^{(k)} < \epsilon$ or $k = K$

8: **return** $V^{(k)}$.

◀ Back to numerical solution